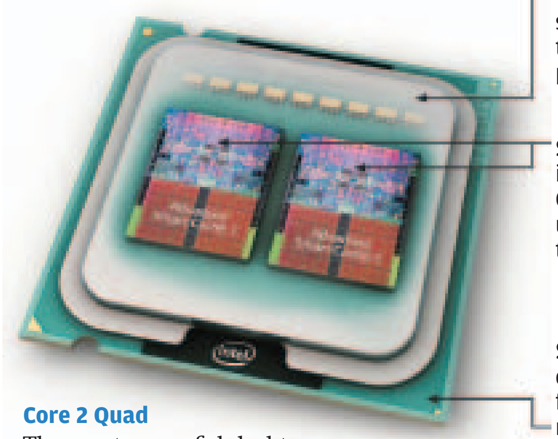
CPUs

Number of Cores

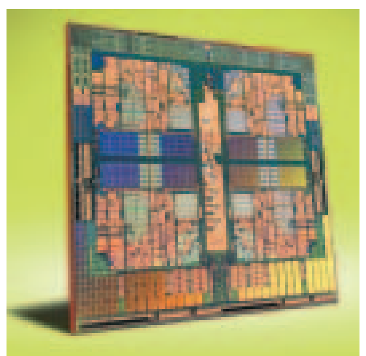


Integrated Heat Spreader (IHS): The integrated metal heat spreader spreads heat from the silicon chips therefore aiding dissipation and protects the die from damage

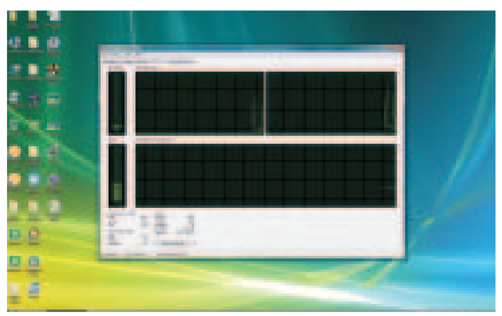
Silicon chips (dies): The two dies inside the Intel Core 2 quad-core CPU are 143mm squared in size and utilise a whopping 291 million transistors each.

Substrate: The dies are mounted directly to the substrate which facilitates the contact to the motherboard and chipset of the PC via 775 contacts and electrical connections.

Core 2 Quad
The most powerful desktop processors around; with the exception of Nehalem of course



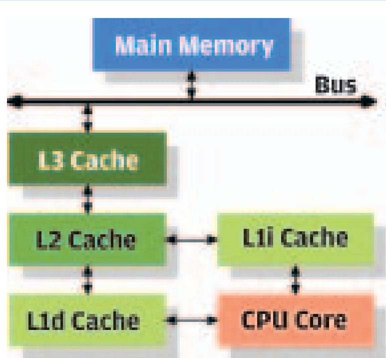
AMD Phenom
Native quad cores, the Phenom processors are amazing value for money

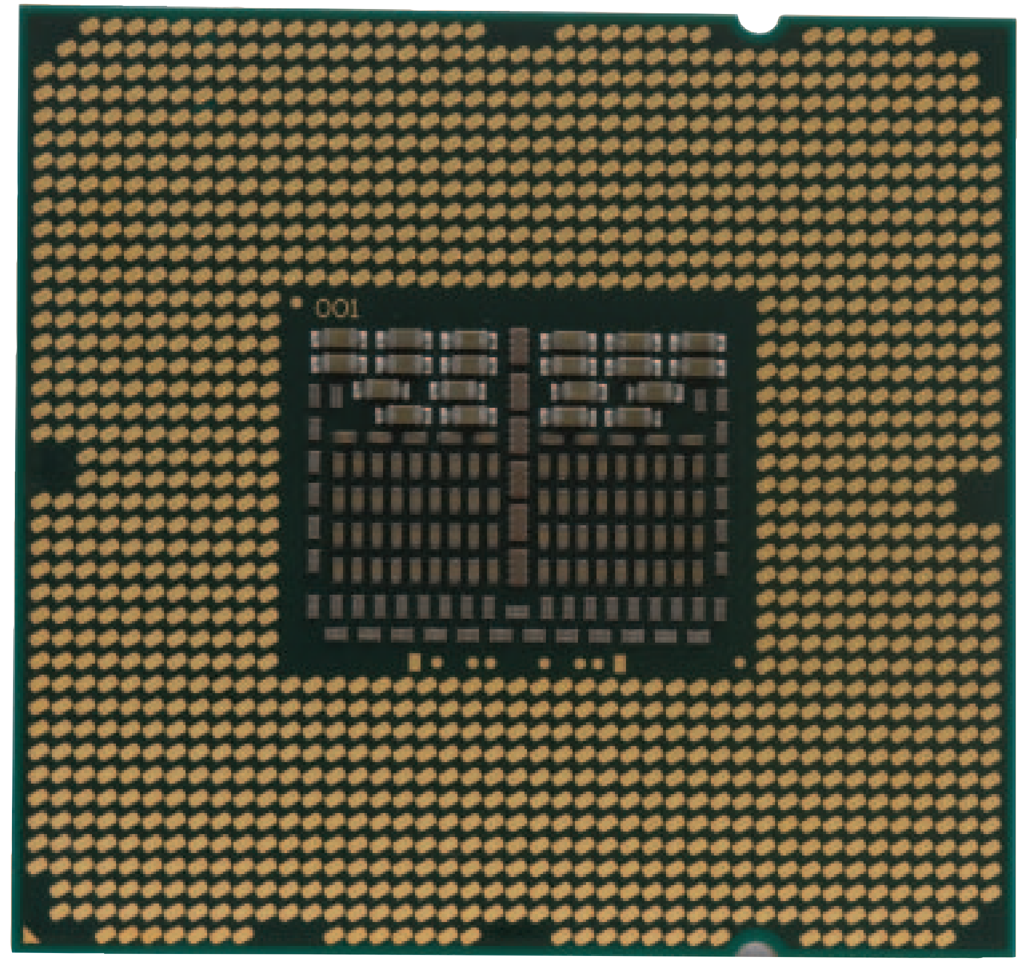


Windows Task Manager
The windows task manager clearly shows how many processing cores your CPU has. There will be four individual panes for a quad core, eight for an octo-core and so on...

Cache

A processors cache is an important part of its performance. CPU Cache is a small amount of storage space that is embedded on the CPU itself to speed up CPU operations by saving it the time to look for the information in main memory. CPUs have L1 and L2 caches. Some may have an L3 cache as well.



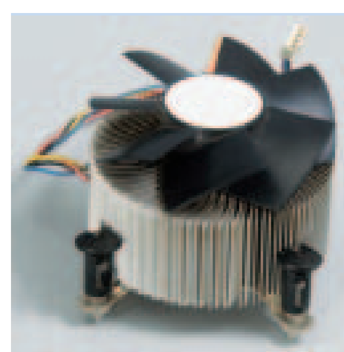


Processors are, by far, the most hyped (sometimes admittedly over-hyped) components in any desktop PC or notebook. Clock speeds are on the rise, and coupled with shrinking fabrication processes desktop and laptop computing has become way more powerful than ever before.

Today Intel builds CPUs on a 45nm process while AMD is still sticking to their older, tested 65nm process. AMD plans to move to 45nm soon and we're told this move will begin with their upcoming processors codenamed Deneb. Intel's latest dual cores codenamed Wolfdale and their quad cores which are codenamed Yorkfield will eventually be phased out in lieu of Nehalem, which is an entirely new architecture for them. However, this does not mean that the older Core 2 Duo and Core 2 Quad processors are useless; price reductions will make them even better value for money and everyone who hasn't jumped on the bandwagon for a new processor may well get an irresistible deal.

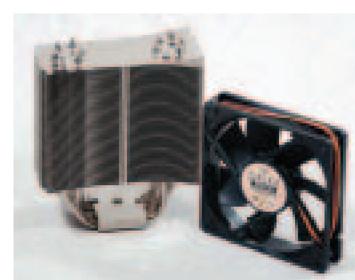
Cooling Solutions

Add-on cooling is a must for some processors that run hot or for overclockers who run their CPUs at higher clock speeds and feed them with higher voltages



Intel's stock cooling solution. Its effective; but the retention is horrible

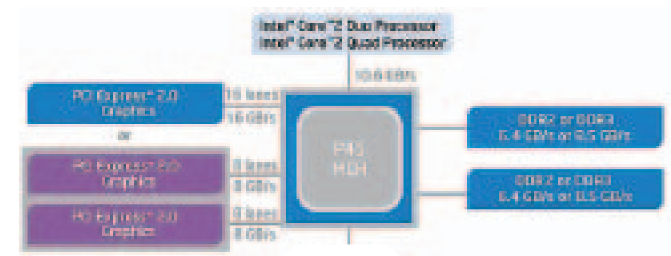
Basic heat-pipe solutions are essential for those looking to run their systems for hours on end



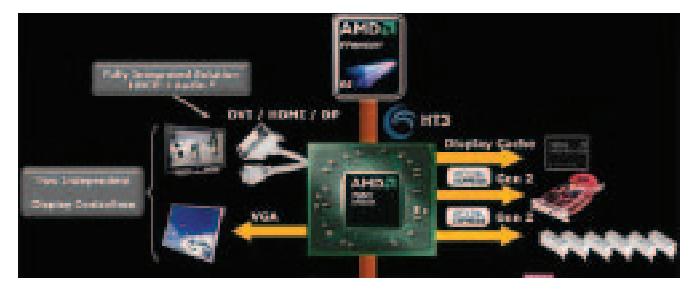
Heatinks like this Thermalright Ultra 120 Extreme are for enthusiasts who are into serious overclocking

Integrated Memory Controller

The memory controller is a logic chip on a motherboard that controls the flow of data to and from the main memory or RAM. Traditionally the memory controller communicates with both the CPU and the RAM via the Front Side Bus or FSB. This FSB is an expressway with 2-way lanes for data travel. However this expressway is used by the CPU to communicate with other components too; and is prone to data congestion; which leads to bottlenecking as the CPU waits for data to reach it. Moving the memory controller to the CPU die itself is the solution as it eliminates the time taken for the FSB to respond to CPU requests.



Intel
Intel has stuck with the concept of an FSB; which is a high speed interconnect between CPU, Northbridge, memory and GPU



AMD
AMD has innovated with developing the concept of a high speed point to point link called Hyper Transport